# Furkan Danisman

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## Education

**Doctor of Philosophy in Statistics** 09/2025 –
*University of Toronto — Co-supervised by Dehan Kong and Zhijing Jin* *Toronto, Canada*

**Bachelor of Science in Statistics** 01/2023 – 05/2024
*York University — First Class With Distinction — GPA: 3.76/4* *Toronto, Canada*

## Awards and Funding

**Canada Graduate Research Scholarship — $27,000** 09/2026 – 08/2027
*Natural Sciences and Engineering Research Council of Canada (NSERC)* *Canada*

**2026 Biostatistics Section Award — Best Student Research Presentation — $500** 06/2026
*Statistical Society of Canada (SSC), 2026 Annual Meeting* *Canada*

**SGS Conference / Research Travel Grant — $480** 2026
*School of Graduate Studies, University of Toronto* *Toronto, Canada*

## Work Experience

**Graduate Researcher** 09/2025 – Present
*Vector Institute for Artificial Intelligence* *Toronto, Canada*

- Working on causal foundational models with amortized Bayesian inference and causal inference in NLP and LLMs.

**Teaching Assistant** 09/2025 – Present
*Statistical Sciences, University of Toronto* *Toronto, Canada*

- Leading tutorials for STA313 (Data Visualization) and STA238 (Statistics and Data Analysis II).

**Research Assistant** 01/2025 – 09/2025
*Banking and Payment Department, Bank of Canada* *Ottawa, Canada*

- **Co-authored** research paper on Decentralized Finance Lending platforms, analyzing trustless borrowing systems where neither party needs to trust each other. Empirically examined trade-offs, functionality, and risks.
- Following publication, co-authored DeFi paper received a **huge media attention** with more than **300,000** views, **40+** news/blogs, 100+reposts, **2 podcast** invite.
- Built Large Value Payment System simulation integrating LLMs as strategic fund managers in an interactive environment. Evaluated how advanced reasoning capabilities improve system efficiency and identified limitations.

**Research Associate | *R*** 05/2024 – 02/2025
*York University* *Toronto, Canada*

- Developed MALC, a bandwidth-free non-parametric density estimation algorithm, for analyzing data with frequencies in interval ranges (e.g., "50 people earn $100k-$120k")
- MALC recovers the underlying density with only a shape-constrained assumption, providing both higher accuracy and a robust option.
- Proved new theoretical convergence properties for EM algorithm with log-concave distributions.

**Research-Boursier** 03/2024 – 10/2024
*The Scientific and Technological Research Council of Turkey* *Ankara, Turkey*

- Developed a two-phase bayesian-frequentist statistical model for longitudinal data with ordinal response.
- Using the model, analyzed 5-8 years of mammographic density follow-ups, demonstrated that longitudinal changes in breast density significantly impact cancer risk.
- Published research in **Scientific Reports**, a multidisciplinary journal by **Nature** Portfolio.

## Research Contributions

### Publications

[1] Chiu, J., & **Danisman, F.** (2026). *DeFi Lending: Returns, leverage, and liquidation risks.* Bank of Canada. Staff Analytical Paper. ***(Cited by 3)***

[2] Erul, E., Pacheco-Barcia, V., Alkan, A., Akkus, E., **Danisman, F.**, Nixon, I., Eniu, A., & Urun, Y. (2026). *Imposter syndrome, burnout, and maladaptive perfectionism among oncology professionals: A global cross-sectional study.* JCO Oncology Practice. ***(Cited by 3)***

[3] Erul, E.*, Avcı, A.N., Akkus, E., Ayas, Ö.F., **Danisman, F.B.**, & Utkan, G. (2025). *Assessment of fear of cancer recurrence in patients with colorectal cancer and its association with pet ownership: A cross-sectional study.* Current Oncology, 32(11), 592. ***(Cited by 2)***

[4] Kulturoglu, M.O., Aydin, F., Sagdic, M.F., Aslan, F., Oflaz, Z., **Danisman, F.B.**, Kalaylioglu, Z., & Dogan, L. (2025). *The impact of changes in breast density over time on breast cancer risk.* Scientific Reports, 15, 23900. ***(Cited by 5)***

[5] Erul, E., Aktekin, Y., **Danisman, F.B.**, et al. (2025). *Perceptions, attitudes, and concerns on artificial intelligence applications in patients with cancer.* Cancer Control, 32, 1–9. ***(Cited by 36)***

[6] Acharya, S., Zhang, T.J., Kim, A., Haghighat, A., Xianlin, S., Shrestha, R.B., Mordig, M., **Danisman, F.**, Chiu, J., Qi, Y.C., Schölkopf, B., Sachan, M., & Jin, Z. (2026). *CauSciBench: Can LLMs automate causal inference in real-world scientific research?* ICML 2026. ***(Cited by 10)***

[7] **Danisman, F.**, Kilic, S.I., Amini, N., Orcun Ada, O., Aktas, S.G., & Sarul, G. (2023). *METU students' college life satisfaction.* International Journal of Social Science Research and Review, 6(7), 12–37. ***(Cited by 1)***

### Under Review

[1] **Danisman, F.***, Zhang, L., Jin, Z., & Kong, D. (2026). *Joint Supervision of Potential Outcomes for Causal Foundation Models.* ICLR 2027.

[2] Chiu, J., & **Danisman, F.** (2026). *DeFi Lending: Returns, leverage, and liquidation risks.* Journal of Financial Market Infrastructures.

[3] **Danisman, F.***, Oflaz, Z., Kalaylioglu, Z., Kulturoglu, M.O., & Dogan, L. (2026). *Decoupling risk and masking in mammographic density under irregular follow-up using a latent Markov progression–detection framework.* Biometrical Journal. (arXiv:2607.16793) **Covered by major Turkish news channels** (media coverage).

[4] **Danisman, F.***, Jankowski, H., & de Souza, C.P.E. (2026). *Bandwidth-free nonparametric density estimation for grouped data.* Computational Statistics and Data Analysis. (arXiv:2607.13182)

[5] **Danisman, F.***, Desai, A., Jiang, J., Rivadeneyra, F., & Azad, T. (2026). *Strategic Interaction Among AI Agents: Evidence from Payment Systems.* Management Science.

### Other Peer-Reviewed Contributions

[1] **Danisman, F.***, Oflaz, Z., & Kalaylioglu, Z. (2026). *Decoupling risk and masking in mammographic density under irregular follow-up using a latent Markov progression–detection framework.* Presentation at the SSC 2026 Annual Meeting Conference (national). **Awarded the 2026 Biostatistics Section Award for the best student research presentation.**

[2] **Danisman, F.***, Jankowski, H., de Souza, C., da Cruz, A., & Rakusin, R. (2025). *Predicting New Onset Atrial Fibrillation: Black-box and Interpretable Models.* Poster presented at the SSC 2025 Annual Meeting Conference (national).

[3] **Danisman, F.***, Nguyen, N., Raghuvanshi, A., & Bisnath, S. (2024). *Augmentation of a global navigation satellite system (GNSS)*. Poster presented at STEM Fellowship Conference (national).

[4] **Danisman, F.***, Jankowski, H., & de Souza, C.P.E. (2025). *Dens-OLog: Density estimation with optimized log-concavity*. Presentation at the SSC 2025 Annual Meeting Conference (national).

**Working Papers**

[1] Arif, S., **Danisman, F.***, Kong, D., Jin, Z., & Mihalcea, R. (2026). *The Culture Within: Towards Mechanistic Understanding of Cultural Reasoning in LLMs*. In preparation for **AAA October** submission.

[2] **Danisman, F.***, Inan, E., Kong, D., & Jin, Z. (2026). *Causal Stories: Representation of the Common Underlying Causal Events*.